TTRTECHNOLOGY

SECURITY OPERATION CENTRE

Inhouse vs. Outsource





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Building a SOC

Establishing a Security Operations Center (SOC) is a complex and resource-intensive initiative that requires continuous staffing, training, and operational support. To effectively monitor, analyze, and respond to cybersecurity threats around the clock, a SOC must be equipped with:

- A secure, dedicated facility
- A robust suite of security and remediation technologies
- Skilled analysts and incident response experts
- Access to real-time threat intelligence

In-house SOCs often operate with limited visibility into the broader threat landscape, relying only on the incidents they've encountered firsthand. To bridge this gap, organizations must subscribe to reliable threat intelligence services that provide insights into both current and emerging risks.

Given how rapidly cyber threats evolve, SOC staff need ongoing training—both internally and through external programs—to stay up to date with the latest security tools and defensive techniques. While collaboration across SOC teams is critical, knowledge sharing is often inconsistent. Talent retention is another major challenge: many skilled professionals leave for opportunities in government or the private sector, where they can work alongside top experts and gain broader experience.

According to Fortinet's 2022 Cybersecurity Skills Gap report:

- SOC roles rank as the second most difficult area to recruit for
- 52% of organizations struggle to retain cybersecurity talent
- 88% of companies report an urgent need to hire cybersecurity professionals

Despite a slight improvement in hiring trends compared to past years, the problem persists. Jon Oltsik, ESG Fellow and cybersecurity expert, emphasizes that the skills gap is not improving meaningfully. In response, organizations are experimenting with automation, expanding their use of managed services, investing in team training, and consolidating security technologies to ease the burden.

The volatility of cybersecurity workforce projections was evident during the COVID-19 pandemic. While Cybersecurity Ventures predicted 3.5 million unfilled jobs by 2021, actual demand in 2020 fell due to pandemic-driven slowdowns. Nonetheless, the long-term outlook remains concerning. According to (ISC)², the cyber workforce gap dropped from 4.07 million in 2019 to 3.12 million in 2020, thanks to a surge of 700,000 new entrants. Still, demand is climbing again—and by 2025, cybercrime is expected to cost the global economy $10.5 trillion, up from $3 trillion in 2015.

Outsourcing a Security Operations Center (SOC)

Outsourcing a SOC provides a cost-effective alternative to building and managing one in-house. It reduces the expense of SIEM license ownership, eliminates the need to hire and train a dedicated internal team, and allows organizations to leverage external expertise and infrastructure.

Key Benefits of Outsourcing a SOC:

24/7 Expert Coverage: Outsourced SOCs are fully staffed around the clock by experienced security analysts. These teams continuously monitor environments, analyze alerts, and coordinate remediation using proven platforms and threat correlation tools.

Threat Intelligence at Scale: Managed SOC providers benefit from global threat intelligence gathered across thousands of client environments. When a new threat is identified in one network, countermeasures are quickly developed and deployed across all customers. This shared intelligence gives outsourced SOCs a major advantage over internal teams with limited visibility.

Rapid Threat Containment: Many outsourced SOCs offer proactive threat removal services, reducing the need for organizations to employ their own incident response (IR) staff. By quickly neutralizing threats, outsourced providers help minimize disruption and damage.

Better Preparedness: When a threat first appears in a local network, global SOCs often already have detection and prevention strategies in place—dramatically reducing exposure time.

Despite these advantages, it's important to choose the right partner. Not all Managed Security Service Providers (MSSPs) offer full incident response. Some simply notify clients of threats and then charge extra for remediation services. This delay in response can increase damage and business disruption, especially if the client lacks the in-house expertise to respond effectively.

Why Speed Matters:

The faster a breach is detected and contained, the lower the damage—both financially and reputationally. According to the 2020 IBM/Ponemon Cost of a Data Breach Report, it took an average of 207 days to identify and 73 days to contain a breach in 2020, totaling 280 days—a slight improvement over 2019. Organizations that detect and respond quickly incur significantly lower breach costs.

Further, FireEye and Mandiant's M-Trends 2021 report notes that the average dwell time—the period a threat remains undetected—dropped to 24 days in 2020, compared to 56 days in 2019 and 416 days in 2011. However, ransomware attacks surged during that same time, rising from 14% of incidents in 2019 to 25% in 2020, highlighting the growing speed and impact of modern threats.

The Bottom Line:

A SOC's effectiveness is determined by its people, tools, and response processes. Outsourcing offers a scalable, intelligence-driven, and often faster alternative to building an in-house SOC—especially for organizations lacking deep cybersecurity expertise. But to fully benefit, companies must choose partners that not only alert but also act swiftly to remediate threats across hybrid, cloud, and on-premises environments.

SOC Analysts: Roles and Responsibilities

According to the SANS Institute, a global leader in cybersecurity training and certification, an effective Security Operations Center (SOC) team should include at least four distinct roles: Tier 1 Analysts, Tier 2 Incident Responders, Tier 3 Threat Hunters, and a SOC Manager (Tier 4). Each role contributes to maintaining constant vigilance over the security landscape of an organization.

Tier 1: Security Analysts

Tier 1 analysts serve as the first line of defense. They continuously monitor network, server, endpoint, database, and application activity for signs of anomalies or threats. These analysts must be skilled at interpreting security logs and identifying suspicious behavior before it escalates into a breach.

Key Tasks:

- Monitor alerts and triage incidents.
- Investigate logs for indicators of compromise.
- Identify system vulnerabilities and unusual activity.
- Escalate confirmed threats to Tier 2 responders

Accurate threat identification is a challenge, especially for organizations without access to comprehensive threat intelligence. Misclassification of threats can lead to inadequate responses, as businesses may not understand the nature, intent, or impact of the attack.

Tier 2: Incident Responders

When a confirmed threat is identified, Tier 1 analysts escalate it to the Tier 2 team. Incident responders perform in-depth investigations, determine the scope of the compromise, and work to contain and eliminate the threat.

Key Tasks:

- Analyze attack data and identify impacted systems.
- Contain and remove threats from the environment.
- Coordinate with analysts to ensure information sharing.
- Conduct forensic investigations and document findings.

Incident response is often one of the most complex areas of cybersecurity. Many organizations mistakenly believe a threat has been neutralized, when in reality, remnants such as backdoors may remain. Trained responders with digital forensics expertise are essential for thoroughly resolving incidents and recommending steps to prevent recurrence.

Tier 3: Threat Hunters

Tier 3 analysts, or threat hunters, are seasoned professionals skilled in proactively seeking out advanced, hidden threats that evade traditional detection methods. They use deep knowledge of networks, forensics, and malware analysis, combined with real-time threat intelligence and indicators of compromise.

Key Tasks:

- Proactively search for undetected threats.
- Analyze sophisticated attacks and reverse-engineer malware.
- Use advanced tools to scan for stealthy intrusion patterns.
- Provide insights to strengthen detection strategies.

Tier 4: SOC Manager

The SOC Manager oversees the entire operation—managing people, processes, and technologies. This role is responsible for recruiting, training, budgeting, scheduling, and ensuring the team operates efficiently. The manager also leads critical incident responses and drives strategic initiatives.

Key Tasks:

- Manage team performance and development.
- Coordinate with business units and executives.

- Oversee SOC budget and vendor relationships.
- Implement new technologies and processes.
- Ensure playbooks and response protocols are up to date.

Conclusion

A well-structured SOC is more than just technology—it's a collaborative team guided by a clear operational playbook. Each tier plays a vital role in identifying, investigating, and responding to threats. With increasing sophistication in cyberattacks, having a fully staffed, well-managed SOC is critical for maintaining security resilience and minimizing risk across all organizational assets.

SOC PROCESSES

SOCs must have established systems in place to ensure that all necessary actions are taken to effectively prevent, detect, and remediate breaches. This includes maintaining the latest intrusion detection and prevention technologies and employing highly qualified personnel.
The SOC should have playbooks that guide the detection and response to threats without disrupting business operations. Routine tasks such as email filtering, network traffic analysis, endpoint monitoring, and client engagement after an incident should be part of the standard operations.

A standardized, repeatable workflow is essential to handle any type of situation, including actions required to meet compliance with regulations such as SOX, FERPA, FISMA, PCI DSS, GDPR, and HIPAA.

To save businesses from spending hours on report creation, the SOC should provide support in meeting compliance requirements for each regulation. It should also offer customized, audit-ready Attestations of Compliance for every client.

SOC TECHNOLOGY

To effectively detect and eliminate threats, SOCs must leverage cutting-edge technologies, including machine learning (ML) and artificial intelligence (AI). With constant data streams flowing from workstations, routers, servers, mobile devices, and various security tools, it's impossible for analysts to process everything manually. Machine learning handles tasks in seconds that would otherwise take hours, helping identify anomalies instantly.

For example, ML can detect unusual behaviors like a U.S.-based employee logging in from a Chinese IP address, which could signal a potential breach. It can also assist in fraud detection by flagging suspicious emails—such as those coming from lookalike domains (e.g., identifying Amazzon.com instead of Amazon.com).

AI builds on this by using machine learning to identify and prioritize threats based on their severity, enabling quicker and more accurate responses.

Furthermore, SOC tools must be capable of monitoring threats both on-premises and in the cloud. Nearly every organization has some form of cloud exposure, even if they're unaware of it. Employees may use cloud-based tools like Salesforce, Dropbox, or Google Docs, or they might be transmitting business data through personal email accounts—creating risks that must be monitored and managed.

COSTS

Companies looking to build a SOC must plan for both initial investment and ongoing operational costs. Before committing, it’s wise to compare the expenses and benefits of building an in-house SOC versus outsourcing it.

In the U.S., the average annual salary for key SOC roles—incident responders, threat hunters, SOC managers, and cybersecurity analysts—exceeds $80,000. Even the smallest organizations need at least one security analyst on duty at all times. In practice, a small SOC team typically requires seven full-time staff: four analysts, one incident responder, one threat hunter, and one SOC manager.

The cost of a Security Information and Event Management (SIEM) system alone averages $50,000, not including annual maintenance and support fees. Add to that the detection and monitoring equipment, which can cost around $610,000, and the financial burden quickly escalates.

“Building a SOC—or generally developing some kind of internal security operations capabilities—is a costly and time-consuming exercise that requires continual attention to be effective,”
— **Siddharth Deshpande, Principal Research Analyst at Gartner**

Because of these high costs, many companies—including some large enterprises—opt against building an internal SOC. Instead, they rely on Managed Security Service Providers (MSSPs) for threat detection and response, which can be more cost-effective and scalable.

OUTSOURCING SECURITY OPERATIONS CENTER

Outsourcing a SOC allows businesses to avoid the cost and complexity of purchasing additional security tools, hiring specialized staff, and investing in continuous training. Instead, they gain access to a fully equipped team of cybersecurity experts who ensure that existing security investments perform at their highest potential.

With 24/7 monitoring and maintenance, outsourced SOCs maximize the effectiveness of a company's current security products. Skilled analysts carefully review all alerts, interpreting log data accurately, while incident responders use their real-world experience to recommend actionable remediation steps.

Clients maintain complete visibility and control over their environments through a single pane of glass, enabling them to monitor, manage, and enforce policies across their infrastructure—all without the operational burden of running a SOC in-house.

CONCLUSION

Operating a SOC is resource-intensive. It demands highly skilled talent, which is both difficult to find and retain, and well-coordinated teams that can share threat intelligence effectively. A SOC's core role is to stay on constant defense—monitoring, detecting, and responding to threats in real-time.

Before investing in a SOC, companies should carefully assess their security goals and challenges. They need to determine whether an internal SOC, an outsourced SOC, or a hybrid model best fits their needs. Some industries or large enterprises may have the infrastructure, budget, and advanced tools to build and maintain their own in-house SOC.

However, SMBs and resource-constrained organizations can still gain access to top-tier cybersecurity protection—without the heavy investment in technology or the struggle of hiring and retaining skilled security personnel. Today, even small businesses can benefit from a 24/7 outsourced SOC, delivering enterprise-grade protection with flexibility and efficiency.

COMMON CUSTOMER CHALLENGES AND HOW WE SOLVE THEM

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Important Factors	Common Challenges Faced while Building your own SOC	How to overcome Challenges within minutes by outsouring SOC as a Service from TTR Technology?
	Security Team	
Finding and recruiting talented applicants	SOC experts are hard to find and harder to keep.	Within a click, have rapid access to industry leading security experts without the burden of recruiting, employing, and retaining internally.
Filling Cybersecurity Skills Gap	Filling gap is challenging for nearly 80% of organizations	
	Costs	
Licensing fees	Costs hundreds of thousands of dollars	SOC as a service works on Monthly Subscription model. You Pay based on Consumption only. No Lock-in Contracts. $0 Onboarding and Transition.
Total Cost of Ownership	Deploying, maintaining, and operating in-house SOC is expensive.	
	Time	
24/7 Monitoring	Hackers attack usually out of working hours	TTR Technology 24x7 Security Team work around the clock
Dwell Time and Economical Impact	Average industry Dwell time is in months	Reduce dwell time from months to minutes, lowering the financial effect
Build a SOC from scratch	Take years to mature SOC processes and scale operations	Within Minutes! JustCall us at 1.855.684.1313
	Tools and Process	
Facilities and Tools	You need to purchase, install, run and maintain all SOC tools	We've all the updated facilities, tools and knowledge to do the job perfectly.
Up-to-Date Security	Keeping up-to-date with the latest SOC tools and capabilities is difficult	
Faster detection and remediation Managed security Events, Alerts, and Incidents Incident & event reporting Threat Detection Threat Detection Vulnerability assessment Root cause analysis Incident Response	You create and manage these processes at your organization instead of focusing on your core business	Our team take away all your worries to let you focus on your revenue growth. Just within a call, you can get the capabilities of a modernSOC without the cost and headache of managing one.
	Tools and Process	
Compliance & Certification	A SOC must be aligned with ISO 27001 or SOC II Type 2.	We're both ISO 27001 or SOC II Type 2 certified

ABOUT

TTR TECHNOLOGY



TTR Technology’s Security Operation Centre helps monitor, detect, and respond to threats in real time, ensuring your systems stay secure.
Contact us today and protect your business with a proactive security monitoring solution.





MISSION

TTR Technology provides managed IT services and cybersecurity solutions that protect your data, optimize business operations, and empower you to grow with confidence.

SOLUTIONS

- Innovation
- Support
- Reliability
- Scalability

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